Power BI Basics

Assignment Solutions

Question 1: What is Power BI and why is it used in businesses?

Answer 1:

Power BI is a business intelligence solution developed by Microsoft. It enables users to connect to different data sources, convert raw data into valuable insights, and create interactive dashboards and reports. Power BI is available in three forms: as a desktop application (Power BI Desktop), a cloud service (Power BI Service), and a mobile app.

Power BI is widely adopted in the corporate world for the following reasons:

- **Data Consolidation:** It supports the connection to hundreds of data sources (Excel, SQL, cloud services, and more) and allows data consolidation from all these sources in one place for comprehensive analysis.
- **Real-Time Dashboards:** Organizations can track KPIs and metrics in real-time through live dashboards, allowing for instant decision-making.
- **Interactive Reporting:** Users without technical knowledge can analyze data through interactive visualizations without writing code or SQL statements.
- **Cost-Effective Solution:** Power BI is available for free in its Desktop version and is very affordable in its Pro versions, making it accessible to all businesses, big or small.
- **Integration with Microsoft Ecosystem:** It integrates seamlessly with Excel, Azure, Teams, and SharePoint, making it an ideal solution for Microsoft-based businesses.
- **Data Governance & Security:** Role-based security and administrative features ensure that sensitive data is accessible only to authorized personnel.

Question 2: Name and explain the three main components of Power BI.

Answer 2:

1. Power BI Desktop: This is a free Windows application that data analysts and developers use to connect to data sources, prepare data by using Power Query, create data models, establish relationships between tables, author DAX (Data Analysis Expressions) measures, and create interactive reports and visualizations. This is the main authoring tool for Power BI.

2. Power BI Service (Power BI Online): This is the cloud-based SaaS offering accessible at app.powerbi.com. The reports and dashboards designed using Power BI Desktop are published to this service for collaboration and sharing. The Service enables data refresh, dashboard creation from pinned visuals, sharing of reports with colleagues, and creation of workspaces for team collaboration. A Power BI Pro or Premium subscription is required for sharing and collaboration.

3. Power BI Mobile: This is the mobile app available for iOS, Android, and Windows platforms. Using this app, users can interact with and explore the dashboards and reports published to the Power BI Service from anywhere. Users can also receive data alerts, mark up reports, and share snapshots of reports with colleagues from their mobile devices.

Question 3: Explain the Power BI workflow.

Answer 3:

The Power BI workflow usually moves through these stages:

- **Step 1 - Connect to Data:** Start by connecting Power BI Desktop to your data. That might mean Excel and CSV files, SQL Server databases, cloud tools like Azure, Salesforce, or Google Analytics, or even web APIs. Power BI has built-in connectors for well over a hundred sources.

- **Step 2 - Transform & Clean Data (Power Query):** With your data loaded, jump into the Power Query Editor. Here's where you clean things up: remove duplicates, filter rows, split columns, fix data types, merge in more tables, or build new calculated fields. Power Query keeps track of each step and reruns them whenever data refreshes.

- **Step 3 - Model the Data:** Once the raw data's cleaner, you load it into the data model. This is where you set up relationships across tables (one-to-many, many-to-many), and create custom columns or DAX measures for your calculations (like Total Revenue = SUM of your sales amount column).

- **Step 4 - Build Visualizations (Report View):** Now you're ready to build charts, maps, tables, or whatever visuals you need on the report canvas. Add filters, slicers, and drill-through actions so reports are interactive.

- **Step 5 - Publish to Power BI Service:** Finished reports get published to the Power BI Service online. Set up data refresh schedules and organize everything into workspaces.

- **Step 6 - Share & Collaborate:** Share reports and dashboards right in the Power BI Service. People can view, play with, and comment on the reports through their browser or the Power BI Mobile app.

Question 4: List any four data cleaning tasks that can be performed in Power Query.

Answer 4:

Power Query Editor makes it easy to tidy up your data. Here are four common tasks:

1. **Removing Duplicates:** It's simple to drop duplicate rows - just pick a column (or columns), then use 'Remove Duplicates' from the Home tab. This keeps your data accurate and avoids double-counting.

2. **Dealing with Missing Values:** When you have empty or null cells, you can fill them in, toss them out, or use options like 'Fill Down' or 'Fill Up' to carry values through. That way, missing data won't throw off your results.

3. **Changing Data Types:** Power Query guesses data types automatically, but it's often a good idea to double-check. If it thinks a date is just text, switch it manually to Date, Number, or whatever fits. This helps with sorting and calculations later on.

4. **Splitting Columns:** Got a column that crams multiple pieces of info together (for example, "Rahul Mehta" in a Full Name column)? Use 'Split Column' in the Transform tab to break it out - maybe into First Name and Last Name - using a space, comma, or whatever delimiter makes sense.

Question 5: Write step by step instructions to load the above dataset.

Answer 5:

Follow these steps to load the Real_Estate_Data_100_Rows.csv file into Power BI:

1. Open Power BI Desktop on your computer. If not installed, download it free from microsoft.com/en-us/power-bi.
2. On the Home ribbon, click 'Get Data'. A dropdown menu will appear with common data source options.
3. Select 'Text/CSV' from the list (since the dataset is a .csv file). This opens the file browser dialog.
4. Navigate to the location where Real_Estate_Data_100_Rows.csv is saved on your computer. Select the file and click 'Open'.
5. A preview window will appear showing the first few rows of the dataset along with detected column names and data types. Verify the data looks correct (delimiter should be comma, encoding should be UTF-8).
6. If the preview looks good, click 'Load' to load data directly into the data model. Alternatively, click 'Transform Data' to open Power Query Editor first for cleaning steps.
7. Once loaded, the table will appear in the Fields pane on the right side of Power BI Desktop with all columns: Property_ID, Property_Name, City, Location_Details, Price, Area_sqft, Bedrooms, Bathrooms, Owner_Name, and Contact.
8. Switch to Data View (table icon on the left sidebar) to browse the loaded data and verify all 100 rows have been imported correctly.

Question 6: Define Data View, Report View, and Model View. Explain the purpose of each view.

Answer 6:

1. Data View: Data View displays the actual data loaded into the Power BI data model in a tabular (spreadsheet-like) format. It shows all rows and columns for each table. Users can use Data View to inspect data after loading, verify transformations applied in Power Query, create calculated columns using DAX formulas, and check data types of each field. It is especially useful for troubleshooting data quality issues. Note: Data View is read-only in the sense that you cannot edit raw data here; changes must be made in Power Query.

2. Report View: Report View is the main canvas where visualizations, charts, graphs, and other report elements are designed and arranged. It is the default view when Power BI Desktop opens. Users drag fields from the Fields pane onto the canvas to create visuals such as bar charts, line charts, pie charts, maps, tables, and cards. Slicers, filters, and cross-highlighting interactions are also configured here. This is the view end-users see when a report is published.

3. Model View: Model View (also called Relationship View) shows all the tables in the data model as boxes and displays the relationships (joins) between them as connecting lines. Users can create, edit, or delete relationships between tables here by dragging fields between tables. Cardinality (one-to-many, many-to-many) and cross-filter direction can be configured. Model View is essential for multi-table data models where data from different sources needs to be related for accurate analysis.

Question 7: Data Sources Supported by Power BI

Answer 7:

Power BI supports a very wide range of data sources, categorized as follows:

1. **File-Based Sources:** Excel workbooks (.xlsx, .xls), CSV/Text files, XML files, JSON files, PDF files, and SharePoint folders. These are the most commonly used sources for small to medium datasets.

2. **Database Sources:** SQL Server, MySQL, PostgreSQL, Oracle Database, IBM DB2, Amazon Redshift, Google BigQuery, Snowflake, SAP HANA, Teradata, and many more relational and analytical databases via ODBC/OLE DB connectors.

3. **Cloud & Online Services:** Microsoft Azure services (Azure SQL, Azure Data Lake, Azure Blob Storage, Synapse Analytics), Microsoft 365 (SharePoint Online, Dynamics 365), Salesforce, Google Analytics, Facebook, GitHub, Zendesk, Stripe, and hundreds of other SaaS platforms.

4. **Power Platform & Microsoft Services:** Power BI Dataflows, Common Data Service (Dataverse), Analysis Services (SSAS/Azure AS), and Power BI Datasets published to the Power BI Service.

5. **Web & API Sources:** Web pages (HTML tables), OData feeds, REST APIs, and ODBC/OLE DB custom connections. Users can also write custom M code in Power Query to connect to virtually any data source.

6. **Other Sources:** R scripts, Python scripts, Hadoop (HDFS), Spark, and Active Directory. This flexibility makes Power BI suitable for any organization regardless of its data infrastructure.

Question 8: Split Owner Name to create two new columns as First Name and Last Name.

Answer 8:

The Owner_Name column in the Real Estate dataset contains full names (e.g., 'Rahul Mehta'). The following steps split it into separate First Name and Last Name columns in Power Query:

Step 1 - Open Power Query Editor: In Power BI Desktop, click 'Transform Data' on the Home ribbon to open Power Query Editor.

Step 2 - Select the Owner_Name Column: In the data preview, click on the 'Owner_Name' column header to select it.

Step 3 - Split the Column: Go to the 'Transform' tab in the ribbon. Click 'Split Column' > 'By Delimiter'. In the dialog, select 'Space' as the delimiter and choose 'Left-most delimiter' to split on the first space. Click OK.

Step 4 - Two New Columns Created: Power Query will create two new columns: Owner_Name.1 (containing the first name, e.g., 'Rahul') and Owner_Name.2 (containing the last name, e.g., 'Mehta').

Step 5 - Rename the Columns: Double-click on 'Owner_Name.1' and rename it to 'First Name'. Double-click on 'Owner_Name.2' and rename it to 'Last Name'.

Step 6 - Close & Apply: Click 'Close & Apply' in the Home tab of Power Query Editor to save the changes and reload the data into Power BI Desktop.